

Shashank Agasimani

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Professional Summary

Forward Deployed Engineer owning the full arc from client requirement to production deployment building voice AI systems, designing structured prompts and conversation flows, integrating into client infrastructure, and driving improvements through evals and A/B testing. Strong technical range across development, RAG, and prompt engineering, with a track record of turning regulated, high stakes workflows into reliable production systems.

Education

KLE Technological University, B.E in Computer Science and Engineering Dec 2020 – Jun 2024

- **Grade:** A
- **Coursework:** Artificial Intelligence and Machine Learning (AIML), Generative AI, AI Agents, MLOPS, OOP, DBMS, Computer Networking, Data Structures, Operating Systems, Web Development.

Experience

Conversational AI Engineer, Arrowhead - Bengaluru Oct 2025 – Present

- Built production-grade, ultra-low latency, human-like AI voice agents using deterministic, state-driven conversation flows and structured prompt engineering for regulated, high-stakes customer interactions across BFSI, NBFC, Insurance, InsurTech, FinTech, WealthTech, Real Estate, and Investment Advisory platforms - achieving upto 45% more conversion rate vs. human agents.
- Integrated voice agents with client-side systems - CRM, telephony, webhooks, internal APIs, and databases - including live human-agent call transfer via client inbound numbers/SIP addresses; debugged, configured, and extended existing infra across multiple live production deployments.
- Evaluated live deployments by auditing calls, logs, and transcripts against baseline conversion/connectivity metrics -diagnosing regressions (objection-handling, drop-offs, latency), running A/B tests across flow versions, and ensuring transcription/translation QA for regional-language deployments.
- Traveled onsite to client locations to conduct requirement gathering, build working relationships with client teams, translating business workflows into deployable conversation flows and fast tracking the delivery.

AI Engineer, AntLabs - Remote May 2024 – Oct 2025

- Led 3+ AI projects end-to-end, owning PRDs, system architecture, hiring, vendor negotiations, and client delivery.
- Designed and deployed a high-precision RAG-based AI chatbot for complex PDF querying, achieving 90–95% accuracy and 65% performance improvement over legacy systems.
- Automated SKU attribute extraction using Gemini API, OCR, and Playwright, reducing processing time by 80%.
- Architected AWS SQS pipeline for real-time image processing, boosting throughput by 5x.

Publications

1. **Panoptic Driving Perception using Deep Learning Techniques** Oct 2023
2. **Vision Based Optical Character Recognition using YOLOv5 and EasyOCR** Apr 2023

Technologies

Voice AI: ElevenLabs, Cartesia, LLM, TTS, STT, VAD, Deepgram, Whisper, OpenRouter, Plivo, Vapi, Exotel, Dialogflow, Pipecat, Pipeshift, SIP, DID, VoIP, Inbound, Outbound

AI Agents & Tooling: RAG, LangChain, LlamaIndex, Ollama

Programming Languages: Python, SQL, C, C++, Java, JavaScript, Rust

Frameworks & Libraries: Django, FastAPI, Git/GitHub

Databases: MySQL, PostgreSQL, MongoDB, ChromaDB, Pinecone

Cloud Platforms & Messaging: AWS, GCP, Azure

Workflow Orchestration & Reliability: Temporal